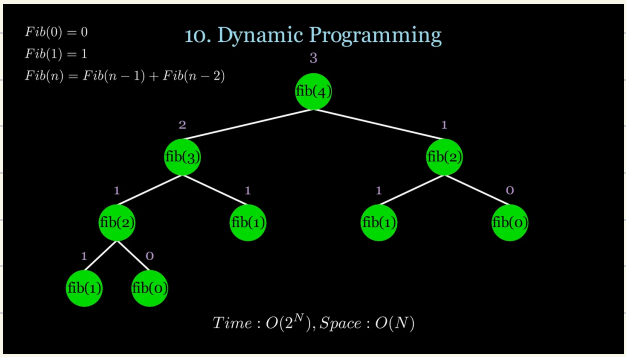
When to use)

- overlapping subproblems and optimal substructure
- optimization problems (min/max distance, profit, etc.)
- sequence problems (longest increasing subsequence)
- combinatorial problems
- reduce time complexity from exponential → polynomial

Types)

- Top Down: you start with the global problem, then recursively split it into subproblems to then solve the global. It is recursion (DFS) with a hashmap that can cache the values previously calculated.

preferred *
- Bottom Up: iteratively preformed by using an array/matrix to store previous values. This method starts w/ base cases then builds up to global solution iteratively.

What is it)

- when you need to solve a problem that depends on prev. results, you can effectively "cache" these previous result values to be reused later. Using two techniques above you can achieve this.